

Linke Song

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Beijing, 100190, China

OBJECTIVE

Interested in computer systems (OS/Virtualization/TEEs). Looking for a system security research internship.

EDUCATION

- **University of Chinese Academy of Sciences, Institute of Information Engineering** Sept.2024 - now
PhD of Computer Systems Organization
Advisor: Prof. Wei Song, Prof. Wenhao Wang
Beijing, China
- **University of Chinese Academy of Sciences, Institute of Information Engineering** Sept.2022 - Jul. 2024
Master of Cyberspace Security
Advisor: Prof. Wenhao Wang
Beijing, China
 - GPA: 3.83/4.00
- **University of Chinese Academy of Sciences** Sept.2018 - Jul. 2022
Bachelor of Cyberspace Security
Advisor: Prof. Wenhao Wang, Prof. Dongdai Lin
Beijing, China
 - GPA: 3.70 (6/20)

RESEARCH PROJECTS

- **NaCRE: Confidential Container with RISC-V Enforcement (WIP)** May. 2025 - now
Tools: C, OpenSBI, Linux Kernel, RunC, Qemu
code
 - Hardware-software co-design for scalable confidential containers on RISC-V — avoiding invasive kernel modifications while preserving security with minimal performance overhead.
 - Building prototype with customized OpenSBI, Linux Kernel, and RunC.
- **LLM sidechannel** Jun. 2024 - Apr. 2025
Tools: Python
code
 - Discovered that KV Cache timing differences in LLM serving systems can leak system prompts; characterized cache hit/miss timing in SGLang with Python scripts.
 - Designed a token-by-token extraction algorithm. **Submission:** Security'25 (RE), CCS'25 (RE), TIFS'25 (AC)
- **NestedSGX** Jul. 2023 - May. 2024
Tools: Rust, C, Python, Linux Kernel Module, Qemu
code
 - Designed an Inter-VM security hardening method; implemented a customized Linux-svsm monitor for NestedSGX Enclave with strong performance and compatibility.
 - Developed Guest Kernel Module and automated performance analysis scripts; received 2 AE badges. **Submission:** ASPLOS'24 (RE), CCS'24 (RE), NDSS'25 (AC)

PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

- [J.1] The Early Bird Catches the Leak: Unveiling Timing Side Channels in LLM Serving Systems.
[Linke Song, Zixuan Pang], Wenhao Wang*, Zihao Wang, XiaoFeng Wang, Hongbo Chen, Wei Song, Yier Jin, Dan Meng, Rui Houu
TIFS' 25
- [C.1] The Road to Trust: Building Enclaves within Confidential VMs.
Wenhao Wang, Linke Song (first student author), Benshan Mei, Shuang Liu, Shijun Zhao, Shoumeng Yan*, XiaoFeng Wang, Dan Meng, Rui Hou
NDSS'25

SKILLS

- **Programming Languages:** C, Java, Rust, Python, lisp
- **DevOps & Version Control:** Git, Docker
- **Specialized Area:** Trusted Execution Environment, Operating System, Virtualization, Container
- **Familiar Architecture:** x86, RISC-V